

J-Verb

Four Japanese studio and broadcast reverbs of the 1980s: dense reflection grids and glassy digital clarity.



Overview

J-Verb brings together four digital reverbs in the spirit of the Japanese studio and broadcast units of the 1980s: dense reflection grids, glassy clarity and clean converters. The PROGRAM keys pick the algorithm, and the whole faceplate changes with it.

HALL is the big modular hall: a dense eight-line network with extra diffusion inside the loop. 3D is a broadcast engine that builds up to 32 reflections from a three-dimensional room model (levels only, so they never cancel in mono) before a high-density tail. FLAGSHIP 40 places up to 40 individual early reflections, then a smooth tail. 12-BIT is the early high-density pioneer: six combs running at a 31.5 kHz clock through 12-bit converters.

Every parameter has its own display on the front panel, as on the hardware remotes: the early reflections and the tail each have their own delay and level, the decay time has separate multipliers for the lows and highs, and the room size sets the scale of everything. A dot-matrix grid shows the reflections as they fall in time.

Reach for it for

- Big, dense halls for pads, strings and ballads (HALL)
- Realistic rooms and stages with strong early reflections (3D, FLAGSHIP 40)
- Grainy, bright 80s plates and drum rooms (12-BIT)
- Long synthwave washes and ambient tails
- Separate control of reflections and tail for placing a sound in a space

Controls

Working the displays

Every parameter has its own display. Drag up or down on a display (Shift for fine moves), use the mouse wheel over it, or press its DOWN / UP keys (hold for repeat). Double-click a display to reset it to its default. The last display you touched is outlined.

HEADROOM

HEADROOM LEDs

Four lights showing the input level at -30, -20, -10 and 0 dB. The 0 dB light is red.

PROGRAM

A HALL

The big modular hall: an eight-line network with allpasses inside the loop for massive density. DRIFT moves its read taps by a sample or two so the tail never sets, without any chorusing. Sparse, wide early reflections.

B 3D

The broadcast engine: up to 32 reflections from a three-dimensional room model (DENSITY sets how many) into a high-density tail. Runs at 48 kHz when VINTAGE is on.

C FLAGSHIP 40

Up to 40 individual early reflections (DENSITY sets how many and how wide they spread), then a smooth tail.

D 12-BIT

Six combs at a 31.5 kHz clock through 12-bit converters, with a few early reflections. The vintage clock is always on for this model; BITS can raise the converters to 16 or 24 bits. Its tail is darker at the top.

VINTAGE CLOCK

Runs the reverb at the unit's own clock rate (44.1 kHz for HALL and FLAGSHIP 40, 48 kHz for 3D, 31.5 kHz for 12-BIT) through converters with the bit depth set by BITS. Always lit on 12-BIT.

BITS

Steps the converter bit depth: 12, 16 or 24. Used whenever the vintage clock is on.

READ PRESETS	Opens the list of factory presets.
Preset display	Shows the last preset recalled (P1 - P10), or P-- if none.
◀ / ▶	The two keys under READ step to the previous or next preset (wrapping round).

The lamp rows

One-touch settings. The lamp lights on the key that matches the current value.

LOW CUT FREQ	Sets LOW CUT to 30, 50, 100 or 200 Hz.
HIGH CUT FREQ	Sets HIGH CUT to 2.5, 5, 10 or 13 kHz.
RT LOW FREQ RANGE	Sets RT LOW to x0.5, x1.0, x1.5 or x2.0.
RT HIGH FREQ RANGE	Sets RT HIGH to x0.25, x0.35, x0.5 or x1.0.

EARLY REFLECTIONS

A dot-matrix picture of the early reflections across 0 - 240 ms: each column of lit dots is a reflection, taller for louder. A faint dotted column marks where the reverb tail starts (PRE DELAY). The grid is empty when E.RFL LEVEL is OFF.

Reverb parameters

REVERB TIME	How long the tail takes to die away at mid frequencies, 0.1 to 20 seconds (2.8 s by default).
PRE DELAY	The delay before the reverb tail starts, 0 to 250 ms (24 ms by default).
REV LEVEL	The level of the reverb tail, -30 to +6 dB; fully down reads OFF.
E.RFL DELAY	The delay before the early reflections start, 0 to 120 ms (8 ms by default).
E.RFL LEVEL	The level of the early reflections, -30 to +6 dB (-6 dB by default); fully down reads OFF.

ROOM SIZE	The size of the space, 10 to 12000 m3 (3000 by default). It scales the reflection pattern and the lengths of the reverb network.
DENSITY	How many early reflections there are (and on FLAGSHIP 40, how widely they spread), 0 to 100%.
DIFFUSION	How quickly the tail smears into a smooth wash, 0 to 100%. Low settings keep more distinct echoes in the onset.
RT HIGH	The decay time above about 5 kHz (3 kHz on 12-BIT) as a multiple of REVERB TIME, x0.1 to x1.5. Below 1 the top end dies away sooner, as in a real room.
RT LOW	The decay time below about 250 Hz as a multiple of REVERB TIME, x0.5 to x2.5. Above 1 the low end rings on longer.
LOW CUT	A high-pass filter on the reverb, 20 to 1000 Hz (80 Hz by default).
HIGH CUT	A low-pass filter on the reverb, 1 to 20 kHz (12 kHz by default).
DRIVE	A clean preamp drive on the reverb input, 0 to 100%: soft saturation before the reverb.
WIDTH	The stereo width of the reverb, 0 to 150%. 0 is mono; above 100 is wider than normal.
DRIFT	How much the reverb wanders over time, 0 to 100% (20% by default). See Drift below.
MIX	The dry / reverb balance, 0 to 100% (30% by default).

Drift

A slow random wander that keeps the tail alive. On HALL it moves the read taps of the network by up to two samples - more density, never a pitch wobble. On the other models it lets the air absorption (the high-frequency damping in the tail) wander. On 12-BIT the old clock also runs very slightly unsteady. The wander speeds up as you turn DRIFT up.

IN / OUT

TRIM	Input level before anything else, -24 to +12 dB. Click the display to select it, then drag, use the mouse wheel or DOWN / UP.
GUARD	Off by default. When on, it gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
LEVEL	The output level, -24 to +12 dB.
SAT	Band saturation on the way out, 0 to 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). The caption reads OUT STAGE - CLIPPING while the clipper is working.

The header

S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Presets set the model and the reverb parameters (including VINTAGE, BITS, WIDTH and MIX). DRIFT and the IN / OUT section are left as they are, except Reset.

Massive Hall	HALL: 4.5 s in a very large room, wide, with longer lows and darker highs.
Vocal Hall	HALL: 2.6 s, a medium hall for vocals, low cut at 100 Hz.

Synth Pad Space	HALL: an 8 s tail in a huge room, very diffuse and wide, 40% mix.
3D Scoring Stage	3D: 2.2 s with dense, strong early reflections.
Film Room	3D: a small 1 s room with prominent reflections.
Studio Room 40	FLAGSHIP 40: a tight 0.7 s room with the full 40 reflections.
Acoustic Guitar (vintage)	FLAGSHIP 40: a 1.3 s room with the vintage clock on.
80s Vocal Plate	12-BIT: a 2 s bright, grainy plate-like reverb at 12 bits.
Drum Room 12-Bit	12-BIT: a short 1 s room with a little drive.
Synthwave Wash	HALL: a 12 s wash in a huge room, vintage clock, 150% width, 45% mix.
Reset all to defaults	Every control back to its starting value, including drift and the IN / OUT section.

Tips

- Use E.RFL LEVEL and REV LEVEL separately: reflections alone place a sound in a room without washing it out; the tail alone gives space without clutter.
- A PRE DELAY of 20 - 60 ms keeps a vocal clear of its own reverb.
- Lower RT HIGH (the x0.35 or x0.5 lamp) for a more natural, darker room; raise RT LOW for warmth.
- On a send, set MIX to 100% and use LEVEL to set the return level.

File: gfx-j-verb.jsfx